

PROBLEMS CLASS 3: SOLUTIONS

1. SIR model revisited

In problems class 1, you might have got far enough to see the popular SIR disease model:

$$\frac{dS}{dt} = -\beta IS, \quad (1)$$

$$\frac{dI}{dt} = \beta IS - \gamma I, \quad (2)$$

$$\frac{dR}{dt} = \gamma I, \quad (3)$$

$$N = S + I + R. \quad (4)$$

We're going to draw a phase portrait of (S, I) . But first, let's remember what this all means.

- (a) What do all these terms represent? Which term is reduced by social distancing? In this model, R is actually composed of two classes of patients: what are they? What is the assumption here about reinfection, and for which diseases is this correct?
- (b) Last time, we introduced the famous reproductive ratio R_0 . What was the intuition behind R_0 ? And what was the definition of R_0 in terms of the parameters we've seen here?
- (c) Consider only the equations for dS/dt and dI/dt and draw a phase portrait for (S, I) .
- (d) Mark the total population, N , onto the S axis. What can you say about the existence of one of your dI/dt nullclines to the left of $S = N$ in relation to the reproductive ratio R_0 ?
- (e) The main reason given for the lockdowns we experienced in 2020–21 was the potential pressure on the NHS, specifically the number of patients requiring hospitalisation at any given time.
 - (i) In our model, what could represent the number of patients requiring hospitalisation at any one time?
 - (ii) At what value of S is this number a maximum?
 - (iii) Divide two of our governing equations to find an expression for dI/dS . Assume that at the beginning of time, I_0 people are infected and the rest of the population is susceptible. Integrate this equation and show that if we assume $I_0 \ll N$, the maximum number of infected patients is given by

$$I_{\max} = N \left(1 - \frac{1 + \log R_0}{R_0} \right). \quad (5)$$

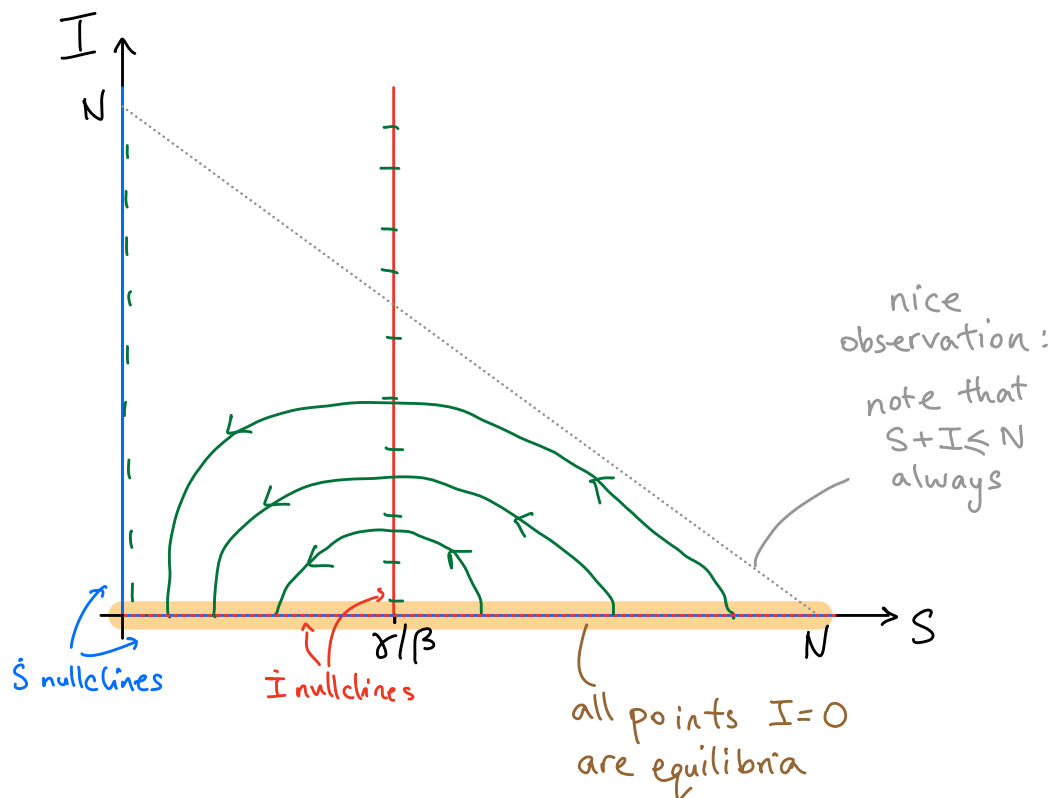
Solution:

- (a) S is the susceptible population, I the infected, R the recovered, N the total population. The term βIS represents the number of susceptible–infected interactions, and γI is the recovery rate.

Social distancing should reduce the value of β , as this represents interactions. This term could be time-dependent: if measuring over days, then lockdown surely would decrease β for a certain period of time.

R includes individuals that have both recovered from the disease, and those who have died. In both cases, they can not catch the disease again.

- (b) We had that $R_0 = \beta N / \gamma$.
(c) The phase plane, if $\gamma / \beta < N$, looks like



Equilibria are all along the line $I = 0$: the first time we've seen this. Equilibria to the left of the $\dot{I}$ nullcline are stable; to the right, unstable.

- (d) First note that S is bounded above by N (you can't have more susceptible people than people). If the $\dot{I}$ nullcline is to the right of N , then I is decreasing in the entire permissible space, i.e. the disease is dying out. What does it mean for the nullcline to be to the right of N ? It means

$$\gamma / \beta > N \implies N / R_0 > N \implies R_0 < 1, \quad (6)$$

which is what we already knew from Problems class 1, but this time we see it graphically.

- (e) (i) With no other information, we could assume that the number of patients requiring hospitalisation is a proportion of those who are infected, I .
(ii) We look for when $dI/dt = 0$, i.e.

$$0 = I(\beta S - \gamma). \quad (7)$$

The maximum (assuming that it is a maximum, of course) occurs when $S = \gamma/\beta$.

(iii) We get

$$\frac{dI}{dS} = \frac{\beta IS - \gamma I}{-\beta IS} = -1 + \frac{\gamma}{\beta S}. \quad (8)$$

Integrating, this gives us

$$I = -S + \frac{\gamma}{\beta} \log S + C, \quad (9)$$

where our initial condition of $I = I_0$, $S = N - I_0$ gives us that

$$C = I_0 + N - I_0 - \frac{\gamma}{\beta} \log(N - I_0), \quad (10)$$

$$= N - \frac{\gamma}{\beta} \log(N - I_0). \quad (11)$$

The maximum occurs when $S = \gamma/\beta$, giving us

$$I_{\max} = -\frac{\gamma}{\beta} + \frac{\gamma}{\beta} \log \frac{\gamma}{\beta} + N - \frac{\gamma}{\beta} \log(N - I_0). \quad (12)$$

Substituting in $R_0 = \beta N/\gamma$,

$$I_{\max} = -\frac{N}{R_0} + \frac{N}{R_0} \log \frac{N}{R_0} + N - \frac{N}{R_0} \log(N - I_0) \quad (13)$$

$$= N \left[1 - \frac{1}{R_0} \left(1 - \log \frac{N/R_0}{N - I_0} \right) \right]. \quad (14)$$

If $I_0 \ll N$, we are left with

$$I_{\max} = N \left[1 - \frac{1 + \log R_0}{R_0} \right]. \quad (15)$$

2. SIR with temporary immunity

So far, we have assumed that immunity is permanent (how has this been encoded into our model?) What if immunity is only temporary? We know that Covid-19 reinfections are possible.

- (a) Introduce a new term to our equations which represents the ‘movement’ of the population from the immune section, R , to the susceptible section, S .
- (b) Show that there are two equilibria of this new model, with one of them given by

$$S = \frac{\gamma}{\beta}, \quad I = \frac{N - \frac{\gamma}{\beta}}{1 + \frac{\gamma}{\delta}}, \quad R = \frac{N - \frac{\gamma}{\beta}}{1 + \frac{\delta}{\gamma}}, \quad (16)$$

where δ is the factor on your R terms in the governing equations. Is this different to our original SIR model, and if so, how?

- (c) By using the equation for N to eliminate R from your expression for dS/dt , write down two equations governing this system.

- (d) By writing down the Jacobian of this reduced system where appropriate, perform a linear stability analysis on the equilibria you have found, and state under what conditions each equilibrium is asymptotically stable/unstable/degenerate. (Hint: you may find the rules about determinants and traces useful.)
- (e) How might you explain this to the general public? (Assuming you are able to crop your slides to the correct aspect ratio.) [This last comment would have been considered humorous and topical in 2020; maybe you remember this...?]

Solution:

- (a) Our new system is

$$\frac{dS}{dt} = -\beta IS + \delta R, \quad (17)$$

$$\frac{dI}{dt} = \beta IS - \gamma I, \quad (18)$$

$$\frac{dR}{dt} = \gamma I - \delta R, \quad (19)$$

$$N = S + I + R. \quad (20)$$

- (b) In looking for equilibria, we set $dS/dt = dI/dt = dR/dt = 0$:

$$0 = -\beta IS + \delta R, \quad (21)$$

$$0 = I(\beta S - \gamma), \quad (22)$$

$$0 = \gamma I - \delta R, \quad (23)$$

$$N = S + I + R. \quad (24)$$

Equation (22) tells us that one possibility is $I = 0$, in which case it follows that $R = 0$ and $S = N$ from the other equations: i.e. this is our disease-free equilibrium.

It also gives us the other possibility that $S = \gamma/\beta$. In which case, replacing R with $N - S - I = N - \gamma/\beta - I$ in eq. (21) gives us

$$0 = -\beta I \frac{\gamma}{\beta} + \delta \left(N - \frac{\gamma}{\beta} - I \right) \quad (25)$$

$$= -I\gamma + \delta \left(N - \frac{\gamma}{\beta} - I \right) \quad (26)$$

$$= -I(\gamma + \delta) + \delta \left(N - \frac{\gamma}{\beta} \right), \quad (27)$$

i.e.

$$I = \frac{\delta(N - \gamma/\beta)}{\gamma + \delta} = \frac{N - \frac{\gamma}{\beta}}{1 + \frac{\gamma}{\delta}}. \quad (28)$$

Then,

$$R = N - S - I \quad (29)$$

$$= N - \frac{\gamma}{\beta} - \frac{\delta(N - \frac{\gamma}{\beta})}{\gamma + \delta} \quad (30)$$

$$= \frac{N(\gamma + \delta) - \frac{\gamma}{\beta}(\gamma + \delta) - \delta(N - \frac{\gamma}{\beta})}{\gamma + \delta} \quad (31)$$

$$= \frac{N\gamma - \gamma^2/\beta}{\gamma + \delta} = \frac{N - \frac{\gamma}{\beta}}{1 + \frac{\delta}{\gamma}}. \quad (32)$$

This is different to our original SIR model because although the disease-free equilibrium is the same, we don't have an additional one-parameter family of equilibria.

(c) As instructed,

$$\frac{dS}{dt} = -\beta IS + \delta(N - S - I), \quad (33)$$

$$\frac{dI}{dt} = \beta IS - \gamma I. \quad (34)$$

The Jacobian, $\mathbf{J}$, is

$$\mathbf{J} = \begin{pmatrix} -(\beta I + \delta) & -(\beta S + \delta) \\ \beta I & \beta S - \gamma \end{pmatrix}. \quad (35)$$

(d) Linear stability analysis: on our disease-free equilibrium, $(S, I, R) = (N, 0, 0)$, our Jacobian is

$$\mathbf{J} = \begin{pmatrix} -\delta & -(\beta N + \delta) \\ 0 & \beta N - \gamma \end{pmatrix}. \quad (36)$$

Since there is a zero on the off-diagonal, the eigenvalues are just the terms on the diagonal, namely

$$\lambda_1 = -\delta, \quad \lambda_2 = \beta N - \gamma. \quad (37)$$

The second eigenvalue is only positive when $\beta N/\gamma > 1$, i.e. $R_0 > 1$, in which case this is *unstable*. If $R_0 < 1$, it is *asymptotically stable*.

[If you did this the trace/determinant way—and you shouldn't, because the eigenvalues can be read off directly—you would get

$$\text{tr}(\mathbf{J}) = \beta N - \gamma - \delta, \quad \det(\mathbf{J}) = -\delta(\beta N - \gamma). \quad (38)$$

The requirement for stability of positive determinant and negative trace means

$$\det(\mathbf{J}) > 0 \implies \beta N - \gamma < 0, \quad (39)$$

$$\text{tr}(\mathbf{J}) < 0 \implies \beta N - \gamma - \delta < 0, \quad (40)$$

but since the bottom condition is weaker than the top, we just have $\beta N - \gamma < 0$, i.e. $R_0 < 1$ for asymptotic stability.]

As for the other equilibrium where $S = \gamma/\beta$, no off-diagonal zeros so we go for the trace/determinant approach directly:

$$\text{tr}(\mathbf{J}) = -(\beta I + \delta) + \beta S - \gamma \quad (41)$$

$$= -(\beta I + \delta) \quad (42)$$

which is negative because β , I and δ are all positive terms. As for the determinant,

$$\det(\mathbf{J}) = -(\beta I + \delta)(\beta S - \gamma) + (\beta S + \delta)\beta I \quad (43)$$

$$= (\gamma + \delta)\beta I \quad (44)$$

which is positive because it consists of purely positive terms. Since stability requires positive determinant and negative trace, this equilibrium is *asymptotically stable*.

(e) (I leave this up to you)

3. Advection and diffusion

Last week we introduced the advection–diffusion equation for a concentration or population $c(\mathbf{x}, t)$,

$$\frac{\partial c}{\partial t} = -\nabla \cdot (c\mathbf{v}) + D\nabla^2 c + f. \quad (45)$$

- (a) In words, what is diffusion?
- (b) In words, what is advection?
- (c) What does f represent?
- (d) What does this equation look like in one spatial dimension, $\mathbf{x} = x$?

Solution: These questions are here just to solidify your intuition.

- (a) (for example) Diffusion is the spreading out of the population over space, essentially due to individual random motion (but the centre of mass stays the same).
- (b) (for example) Advection is the movement of the centre of mass of the population, or in other words a bulk motion (think ducks being carried away down a river).
- (c) $f(c)$ is any source or sink term, and this can be any of the population models we have seen so far this term.
- (d) In one spatial dimension, this is

$$\frac{\partial c}{\partial t} = -\frac{\partial}{\partial x}(cv) + D\frac{\partial^2 c}{\partial x^2} + f. \quad (46)$$